

- 1 (a) By placing **one** tick (✓) in each row, complete Table 1.1 to identify whether each of the listed quantities is a scalar or a vector.

Table 1.1

quantity	scalar	vector
amplitude		
force		
momentum		
power		

[2]

- (b) A car moves in a straight line along a road. Figure 1.1 shows the variation with time t of the velocity v of the car.

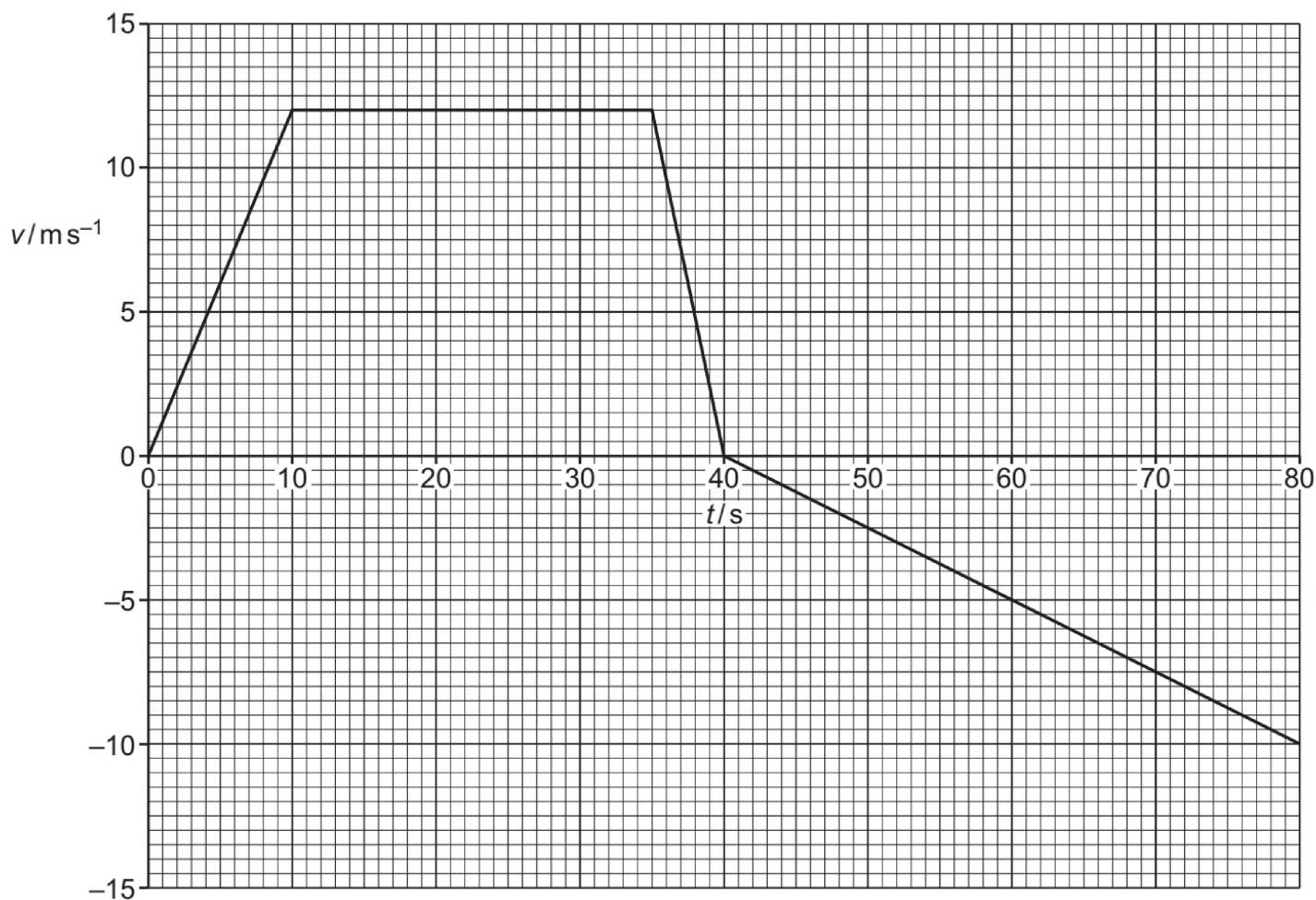


Figure 1.1

- (i) State the feature of the graph that represents the distance travelled by the car between $t = 0$ and $t = 40$ s.

..... [1]

- (ii) Calculate the distance travelled by the car between $t = 0$ and $t = 40$ s.

distance = m [1]

- (iii) Determine the maximum magnitude of the acceleration of the car.

acceleration = ms^{-2} [2]

- (iv) At $t = 25$ s, the car is at a point P.
At $t = 40$ s, the car is at a distance of 150 m from P.
After $t = 40$ s, the car travels with a constant acceleration of -0.25 ms^{-2} until the car is again at point P.

Calculate the speed of the car when it is at point P for the second time.

speed = ms^{-1} [2]

(v) At $t = 0$, the displacement of the car is zero.

On Figure 1.2, sketch the variation of the displacement of the car with t from $t = 0$ to $t = 80$ s. Values for displacement are **not** required.

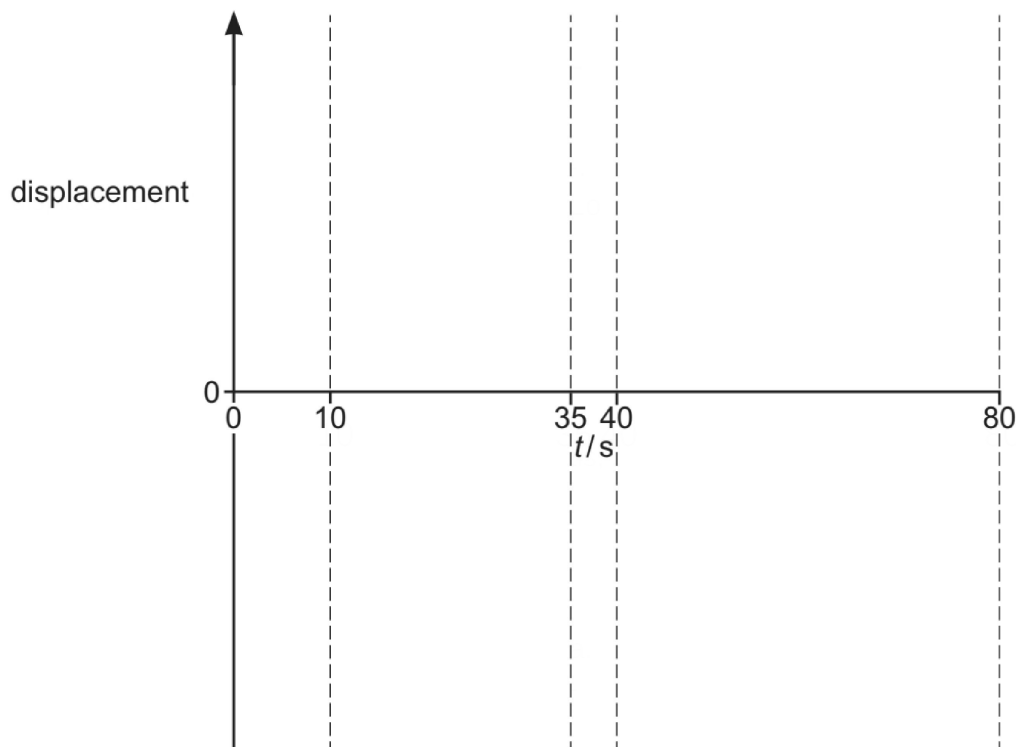


Figure 1.2

[3]

[Total: 11]